

Problem 6.22

On April 1, 2013, you will need \$10,000. Assuming a 6% annual compound effective rate of interest, what would you have to invest on October 1, 2008, in order to fulfill that need?

Solution

From October 1, 2008 to April 1, 2013, the investment will accumulate for 4.5 years.

Let x be the amount invested on October 1, 2008. Then, using a 6% annual effective rate,

$$10000 = x(1.06)^{4.5}.$$

Solving for x ,

$$x = \frac{10000}{(1.06)^{4.5}} \approx 7693.49.$$

Therefore, the amount that must be invested on October 1, 2008 is

$$\boxed{\$7,693.49}$$